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24 September 1987
SUPERSEDING
MIL-S-51422A(EA)
21 June 1979

MILITARY SPECIFICATION

# SHELTER AND ENTRANCE ASSEMBLY - CHEMICAL-BIOLOGICAL SHELTER SYSTEM: INFLATABLE

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for one type and size of semicircular-shaped, double-walled, air-supported chemical-biological (CB) shelter tent assembly, including entrance. It also provides for procurement of either the shelter or entrance separately (see 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| V-F-106 | - | Fasteners, Slide Interlocking. |
| V-T-285 | - | Thread, Polyester. |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard. |
| PPP-C-843 | - | Cushioning Material, Cellulosic. |
| PPP-T-60 | - | Tape: Packaging, Waterproof. |

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-5425 - Plastic, Sheet, Acrylic, Heat Resistant.
MIL-F-21840 - Fastener Tape, Hook, and Pile, Nylon.
MIL-C-43307 - Cord, Nylon, Solid Braid, General Purpose.
MIL-C-43944 - Cloth, Laminated and Cloth, Coated (Chloroprene Coated).

STANDARDS

FEDERAL

Fed. Test Method Std. No. 191 - Textile Test Methods.
Fed. Std. No. 595 - Colors.
Fed. Std. No. 751 - Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-171 - Finishing of Metal and Wood Surfaces.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

U. S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-19-5178 - Shelter, Inflatable, Chemical-Biological, Shelter System.
5-19-5250 - Entrance, Inflatable, Chemical-Biological, Shelter System.
PL 5-19-6822 - Shelter and Entrance with Accessories.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-5178 - Shelter, Inflatable, M51.
P5-19-5250 - Entrance, Protective.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

United States Army Standard Color Card for the Official Standardized Shades of Sewing Threads.

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, New York 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standards

D 1003 - Haze and Luminous Transmittance of Transparent Plastics.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawings 5-19-5178 and 5-19-5250 and the accessory items on PL 5-19-6822 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings. PL 5-19-6822 describes one of two subsystems which comprise the M51 Shelter System.

3.1.1 Shelter fabrics.

3.1.1.1 Exterior walls. The exterior walls of the shelter section shall be fabricated from cloth conforming to type I, class 1, of MIL-C-43944.



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3.1.1.2 Interior walls. The interior walls of the shelter section shall be fabricated from cloth conforming to type I, class 2 of MIL-C-43944.

3.1.1.3 Seal webs. The seal webs at the section junctures shall be fabricated from cloth conforming to type III of MIL-C-43944.

3.1.1.4 Retaining webs. The retaining webs within the section junctures shall be fabricated from cloth conforming to type IV of MIL-C-43944.

3.1.1.5 Floor material. The floor of the shelter section shall be fabricated from cloth conforming to type II of MIL-C-43944.

3.1.2 Tape, chloroprene-backed. The tape used to seal the butt seams and the exposed web-seam stitching on both the exterior and interior of the shelter shall be a polyvinylfluoride film to which is applied an uncured chloroprene rubber, 0.013 (-0 or +0.003) inch thick. This polyvinylfluoride film shall conform to L-P-1040, type III, class 1, grade A, olive-green 207 for the exterior of the shelter and type III, class 1, grade B, white for the interior of the shelter. The uncured chloroprene film tape shall be applied to the seam using type I adhesive, 3.1.8, as indicated in the drawings with the chloroprene side facing the seam and cured. The formulation of the uncured chloroprene, the time lapse from its application to the polyvinylfluoride film to its use in the finished shelter, and the method of curing the chloroprene shall be such that the performance requirements of 3.1.8.1 and 3.3.2.3 are met. Agent penetration resistance shall be not less than 100 minutes for mustard (H) and 200 minutes for GB when tested as specified in 4.4.10.1.

3.1.3 Window material. The material used in the window shall be a clear polyurethane film either a single layer 0.010 $\pm$ .002 inch thick or two layers, each 0.005 inch thick. If a double layer, the film shall be laminated together with either heat and pressure or polyurethane adhesive to form an optically clear laminate 0.010 $\pm$ .002 inch thick. The window material shall conform to the following requirements.

(a) After accelerated weathering in accordance with Method 5804 Fed Test Method Std. No. 191 for 200 hours, the window material shall have a haze factor not exceeding 25 when tested as specified in 4.4.11.

(b) The window material shall maintain flexibility and there shall be no indication of cracking or delamination (if laminated) at -25°F.

(c) The initial and after low temperature agents penetration resistance shall be no less than 100 minutes for Mustard (H) and 200 minutes for GB when tested as specified in 4.4.10.



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3.1.4 Viewing ports, entrance. The material for the two airlock door viewing ports shall be plexiglass, conforming to MIL-P-5425. The windows shall be fabricated from stock of the required thickness and shall be the size specified on Drawing 5-19-5250. Material used for this application shall not be tinted, stained, or otherwise colored.

3.1.5 Thread, polyester. The thread shall conform to type I, class 1, sub-class B of V-T-285.

3.1.5.1 Color. The thread for stitching on the inside of the shelter section shall be white in conformity with color No. 37875 of Fed. Std. No. 595 or shall conform to the applicable shade of the U.S. Army Standard Color Card for the Official Standardized Shades of Sewing Threads. The thread for stitching on the outside of the shelter shall be dyed olive drab shade S-1 (cable No. 66022). The dyed thread shall show fastness to weathering equal to or better than the standard shape sample (see 6.3). If no standard shade sample is available, the dyed thread shall show good fastness to weathering.

3.1.6 Fastener tape. The nylon hook and pile fastener used in the shelter and entrance shall conform to type I, class 1 of MIL-F-21840 except that no color samples shall be required. Colors shall conform to Velcro No. 330 (Black), No. 660 (OD) and No. 12 (White). Maximum width shall be 1.00 inch and minimum width shall be 0.625 inch.

3.1.7 Fastener, slide. The interlocking slide fastener which attaches the removable wall section to the rear shelter wall and is used in shelter interconnection shall conform to B.F. Goodrich type 411, vapor-tight zipper or equal. The slide fastener chain shall also conform to the applicable section of V-F-106.

3.1.8 Adhesives. The adhesives used for shelter fabrication and for joining assemblies shall be of the types listed below (see 6.5):

Type I - Adhesive for Bonding Seam Sealing Tape.
Type II - Adhesive for Bonding Structural Seams.

3.1.8.1 Adhesive requirements. When tested as specified in 4.4.3, type I and II adhesives shall permit no separation, slippage, or edge peeling of either the butt seam strap or the sealing tape.

3.1.9 Lines, tent. The lines shall conform to MIL-C-43307.

3.2 Manufacture and assembly. The shelter and entrance assemblies shall be manufactured as specified on Drawings 5-19-5178 and 5-19-5250 or on subsidiary drawings.

3.3 Design and construction. The design and construction of components and assemblies covered by this specification shall conform in all respects to the drawings listed in section 2 and as specified herein.



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3.3.1 Stitching, machine.

3.3.1.1 Types of stitching. Stitching shall conform to Fed. Std. No. 751, as follows:

For all stitching - Type 301 and Type 401, 5 to 7 stitches per inch.

3.3.1.1.1 Type 301 and type 401 stitching. Thread breaks in stitching shall be backstitched not less than one inch at each break. Stitching ends, except where ends are turned under in a hem or held down by other stitching, will be:

EITHER: backstitched not less than one inch.

OR: cemented, by extending the stitch line not less than one inch beyond the joined material and applying adhesive over the stitching extension so as to hold the thread to the fabric and fill the needle holes.

3.3.1.1.2 Box stitching. Box stitching and WW stitching shall be used alternately depending upon the type of machines available to the contractor. Box stitching and/or WW stitching shall conform to requirements for stitch pattern, stitches per inch, size and type of thread, and at least three or more tying, overlapping or backstitches shall be used to secure the ends of stitching.

3.3.1.2 Type of seam. Seam type shall conform to Fed. Std. No. 751 requirements for the seams shown on the applicable drawings.

3.3.1.3 Thread ends. All thread ends shall be trimmed to not more than one half inch length.

3.3.2 Joints, seams, and splices.

3.3.2.1 Cementing. All joints, seams, and splices to be cemented and sealed shall be thoroughly cleaned and prepared for cementing and shall be smoothly cemented and free of puckers, unadhered pockets, or tunnels. Care shall be exercised, while the adhesive is curing, that the cemented joints, seams, and splices shall not be subjected to handling or draping stresses sufficient to produce slippages and separations.

3.3.2.2 Breaking strength. All cemented structural joints, seams, and splices shall have a minimum cut-strip breaking strength of 200 pounds when tested as specified in 4.4.1. This requirement shall not apply to non-structural cemented joints. Structural joints, seams and splices are any cemented composite which comes under direct stress when the structure is inflated.

3.3.2.3 Dead load strengths at room temperature and at 200°F. Cemented butt seams tested periodically during production of the shelter under condition (a) and condition (b) of 4.4.4.2 shall show no separation, slippage, or edge peeling of either the butt seam strap or of the sealing tape.



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3.3.2.4 Overpressure seam strength. When the shelter and entrance assembly is subjected to the inflation test described in 4.4.5, there shall be no evidence of (a) separation or slippage between the layers composing the cemented butt seams, no evidence of (b) excessive overload in the sewn seams composite resulting in edge peeling, slippage or separation of the sealing tapes throughout the shelter.

3.3.2.5 Sewn web seam strength. In addition to the requirements specified in 3.3.2.2, the shelter material and the material utilized for seal webs when incorporating a sewn juncture shall be capable of resisting separation under dead loads of 50 pounds for 24 hours at 70°F + 2°F and 25 pounds for 24 hours at 180°F ± 2°F when tested as specified in 4.4.2.

3.3.3 Attachment of slide fastners. Care shall be exercised to trim all thread ends and maintain specified distances between edges of fabric attached to slide-fastener tape so that there will be no interference with the operation of the slider.

3.3.4 Holes for inflation ducts and air pumps. Care shall be exercised when cutting the holes in the walls for the supply, return, and inflation air ducts so that the holes shall be smoothly circular. The edges of the cloth shall closely encompass the ducts or sleeves. Aluminum duct rings shall be both fastened and bonded to the shelter material.

3.3.5 Repairs. Major defect to cloth, tape repairs, misplaced needle holes repairs, and scratches or abrations repairs shall be effected as follows:

3.3.5.1 Repairs to cloth.

3.3.5.1.1 No more than six patches shall be allowed per section. A section shall be defined as an inner and outer panel with webs fabricated as a unit.

3.3.5.1.2 Patches shall extend beyond the defect by a minimum of 1-1/2 inches in all directions.

3.3.5.1.3 Patches shall be applied so that the direction of the weave corresponds to the weave of the shelter fabric.

3.3.5.1.4 All patching shall be accomplished using the same adhesive and the same surface preparation employed on the joints, seams, or splices.

3.3.5.1.5 The contracting officer shall be notified of any repairs made on defects in excess of six inches by length.

3.3.5.2 Tape repairs.

3.3.5.2.1 Over-taping shall be applied only when peeling and re-cementing of the original tape is not feasible (i.e. will damage the wall's skin).



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3.3.5.2.2 The tape used for repair shall be the same width of the original tape.

3.3.5.2.3 No more than six patches shall be allowed per section (as defined in 3.3.5.1.1 not including the six patches allowed for the wall skin.

3.3.5.2.4 The mimimum repair tape length shall be 12 inches regardless of the size of the defect.

3.3.5.3 Misplaced needle hole repair.

3.3.5.3.1 Apply a type III material patch to the neoprene side to extend beyond the needle holes by a minimum of 1/2 inch in all directions.

3.3.5.3.2 Apply a gum tape to the tedlar side.

3.3.5.3.2.1 Apply a strip of gum tape extending 1/2 inch width direction beyond both the needle holes and the sewn seam when both are 3/4 inch or less apart (adjust width of tape as necessary).

3.3.5.3.2.2 When the needle holes are 3/4 inch or more from the nearest sewn seam, use additional gum tape of the same width as typical for that panel to cover the needle holes

3.3.5.4 Scratch or abrastions (where base cloth is undamaged).

3.3.5.4.1 Apply gum tape over the scratch or narrow abrasion using tape which is typical for that panel, extending end 1-1/2 inch beyond the ends of the scratched or abraded area.

3.3.5.4.2 Wider gum tape may be used on wider abrasions so as to extend a minimum of 1/2 inch in all directions beyond the abrasion.

3.3.6 Air leakage.

3.3.6.1 Operational leak. Each completed shelter and entrance assembly shall not exceed 3.0 cubic feet per minute (cfm) leakage when tested as specified in 4.4.6.1.

3.3.6.2 Stand-up leak. Each completed shelter and entrance assembly shall be capable of maintaining a minimum internal air pressure of 14 inches water gage (iwg) when tested as specified in 4.4.6.2.

3.3.7 Internal pressurization. The interior pressure shall be not less than 0.80 iwg for the shelter, and not less than 0.30 iwg for the entrance when air is introduced as specified in 4.4.7.

3.4 Preproduction sample. Before production commences, a sample of the item (see 6.2) shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve



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the contractor of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.5 Interchangeability. Entrances and shelters shall be interchangeable and demonstrate performance in accordance with 3.3.7, when tested in accordance with 4.4.8.

3.6 Workmanship. Coated cloth components shall be free of pinholes, thin or weak places caused by abrasion, exposed fabric, blisters, tunnels, or delamination of coating, creases, wrinkles, or coating missing. Webbing and tape shall contain no frayed or scalloped edges. Cords and rope components shall be free of cuts or abrasions. Thread tension shall be maintained so that there will be no loose stitching and seam allowance shall be maintained with seams properly sewn so that no runoffs, twists, pleats, puckers, or open seams shall result. When attaching slide fastener tape to the cloth, specified distances between edges of cloth and the chain shall be maintained so that there will be no interference with the operation of the slider. Care shall be taken in sewing that there shall be no needle chews. All thread ends shall be trimmed. Metal components shall be free of burrs, sharp edges, and corroded areas and shall not be broken or malformed. In addition all welds shall be as specified on the drawings and materials to be welded shall be clean and free from scale, corrosion, oil, water, and other matter. Welds shall be sound, smooth, free from flash, and show good fusion.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government responsibility. The Government will be responsible for performing the H and GB agent resistance tests specified in 4.4.10.1.



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4.2 Preproduction inspection. A preproduction sample of four shelter and entrance assemblies submitted in accordance with 3.4 shall be examined and tested for all requirements of the applicable drawings and this specification. The contractor shall provide to the Government for testing in accordance with 4.4.1 either of the following:

(a) A one-half pint sample of all adhesives (with sufficient accelerator) to be used during production, along with application instructions.

(b) A one yard square sample of material with each type structural seam (as defined by 3.3.2.1) centrally located. The fabrication of the test seam shall use the same equipment and procedures followed during production and shall be witnessed by a Government representative.

4.3 Sample for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 unless otherwise indicated.

(a) Tape, chloroprene-backed. One roll of chloroprene-backed tape from each lot of tape shall be submitted for examination and test. In addition, no more than one lot of polyvinyl fluoride film shall be represented in any one lot of chloroprene-backed tape (see 3.1.2).

(b) Window material. Twenty-five square feet of window material from each lot of window material shall be submitted for examination and test. In addition, no more than one lot of polyurethane film shall be represented in any one lot of window material (see 3.1.3).

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

4.3.2 In-process inspection.

4.3.2.1 In-process examination. Examination shall be made to see that the following are in accordance with specification requirements:

(a) Joints, seams, and splices shall conform to requirements of 3.3.2.1 and shall be visually examined to ensure freedom from puckers, unadhered pockets, or tunnels.

(b) Examination shall ensure that coated cloth surfaces are cleaned and prepared for cementing in conforming with 3.3.2.1.

Whenever a nonconformity is noted, correction shall be made to the process and to the items affected.



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4.3.2.2 Testing. Testing of the cloth joints, seams, and splices for compliance with 3.3.2.2, 3.3.2.3, and 3.3.2.5 shall be performed as specified in 4.4.1, 4.4.4, and 4.4.2 respectively. The sample unit shall be one joint, seam, or splice per shelter for each of the two classes of coated fabric (see 3.1.1). The number of determinations per test shall be one and the results reported as pass or fail.

4.3.3 End product inspection. A lot shall consist of all completely fabricated shelter/entrance assemblies offered for inspection at one time. The sample unit shall be one complete shelter/entrance for purposes of inspection described in 4.3.3.1, 4.3.3.2, and 4.3.3.3. Note that certain inspections must occur in process, as indicated by asterisks in table II, and not during End Item Examination.

4.3.3.1 Visual examination. The completely fabricated shelter/entrance assembly shall be examined for the defects classified in table I.

TABLE I. Classification of defects (visual examination)

| Component Or Process | DEFECT | Classification Major | Classification Minor |
|---|---|---|---|
| Coated cloth | (a) Thin or weak place due to abrasion; exposed cloth, blisters, tunnels, or delamination of coating; creases or wrinkles. | X | |
| | (b) Lumps or heavily-coated edge. | | X |
| Webbing and tape | Frayed or scalloped edge. | | X |
| Hardware | (a) Broken or malformed. | X | |
| | (b) Finish omitted; corroded area; burrs or sharp edges. | | X |
| Fastener, slide | Slider jams or fails to interlock chain; any defect of slide fastener assembly resulting in improper or ineffective functioning of the slide fasteners. | X | |
| Stitched seams | (a) Any open seam on slide fastener assemblies. (Note 1) | X | |
| | (b) Open seam on other than slide fastener assemblies more than 1/4 inch, but not more than 3/4 inch. | | X |
| | More than 3/4 inch. | X | |



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TABLE I. Classification of defects (visual examination) (continued)

| Component Or Process | DEFECT | Classification Major | Classification Minor |
|---|---|---|---|
| | (c) Thread breaks - over-stitched less than one inch on stitch type 301 and 401. | | X |
| | (d) Needle chews. | | X |
| | (e) One or more required rows of stitching omitted, except on box stitching. | X | |
| | (f) Box stitching incomplete: | | |
| | (1) One end row of stitching omitted. | | X |
| | (2) Two or more rows of stitching omitted. | X | |
| Seam and stitch type | Wrong seam or stitch type. | X | |
| Stitch tension | Loose, resulting in an exposed bobbin or top thread; tight, resulting in puckering of fabric on seams; | | |
| | (a) For more than one inch but less than six inches. | | X |
| | (b) Six inches or more. | X | |
| Stitches per inch | (a) One stitch less than minimum specified. | | X |
| | (b) Two or more stitches less than minimum specified. | X | |
| | (c) One stitch in excess of maximum specified. | | X |
| | (d) Two or more stitches in excess of maximum specified. | X | |
| Stitching gage | Not as specified. | X | |
| Stitching margins | Greater than specified by more than 3/16 inch. | | X |



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TABLE I. Classification of defects (visual examination) (continued)

| Component Or Process | DEFECT | Classification Major | Minor |
|---|---|---|---|
| Stitching ends | Overstitched less than one inch (except where ends are held down by other stitching or turned under in a hem). | | X |
| Components and assembly | Any required component or operation omitted (unless otherwise specified herein). | X | |
| Inflation duct | Misplaced, i.e., not cut out on panel as indicated on Drawing No. E5-19-5180. | X | |
| ECU duct | Sleeve opening. | | |
| | (a) Too small to accommodate duct pipe. | X | |
| | (b) Too large, causing excessive looseness or improper fit with duct pipe. | X | |
| Repairs | Not in accordance with specified instructions (see 3.3.5); not authorized by the contracting officer. | X | |
| Cleanliness | Noticeable grease or stains, thread ends not trimmed to 1/2 inch or less. | | X |

Note 1. A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur and are not overstitched.

4.3.3.2 Dimensional examination. Any dimension listed in table II, not within the specified tolerance, shall be classified as a defect.

TABLE II. Dimensional examination

| DRAWING/SHEET #/DIMENSIONS |
|---|
| E5-19-5178 |
| Shelter, Inflatable, Chemical-Biological Shelter System |
| 1 Section AA & BB 0.25 ± .13 |
| Front View - 68.64" |
| 2 Section QQ - 1.5" |
| *View K - 2.00" |



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TABLE II. Dimension examination (continued)

| | |
|---|---|
| E5-19-5179 | |
| Floor, Shelter | |
| Sections BB, CC & DD - 1.50" + .38 - .13 | |
| E5-19-5180 | |
| Wall Assembly, Front LH | |
| 1 | Main View - 83.50" & 61.50" |
| | *View D - 0.44, 0.25 & 0.75 (two places) |
| | *Check web dimensions by calculation |
| 2 | *View KK - Hole diameters, locations and number of holes |
| | *Typical Joint |
| | *Section EE - location item 20 (9.50"), and foldover |
| | *Section FF - 0.25 ± .13 |
| E5-19-5181 | |
| Wall Assembly, Front, RH | |
| 1 | Main View - 83.50" & 61.50" |
| | *View C - 0.44, 0.25 & 0.75 |
| | *Section EE & FF - Web Fold |
| | *Typical Joint |
| | *Calculate Web dimensions |
| 2 | *Section GG & HH, JJ & KK - 0.25" |
| | *Typical web tape termination |
| E5-19-5182 | |
| Wall Assembly, Rear | |
| 1 | *Main View - Web spacing (7.26") |
| | *View F - Calculate web width |
| | *Section GG - 0.25" |
| E5-19-5183 | |
| Arch Assembly, Shelter | |
| 1 | 191.94" and 212.00" |
| 2 | *View G - Tape seam and stitching |
| 3 | *Section LL - All dimensions |
| | *View M - All dimensions |
| | *Section BB - Web fold and warp direction |
| | *Section CC - Warp direction |
| | *Calculate web dimensions |



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E5-19-5184
Entrance Assembly
1 Main View - 143.62"
End view - Mail exhaust fitting location
*warp direction
*Section AA - Calculate web dimensions
2 *Section FF - web fold and warp direction
*View H, M & N - 0.44", 0.25", 0.75", and 0.19"
*View L - Typical butt seam
*Section UU - 0.44 & 0.19
*Detail J - tape, seam and stitching

E5-19-5185
Wall Section, Removable
Section AA - 47.50
Calculate web dimension
*Section BB - warp direction and fold
Main view - start of zipper (2.25")
*Warp direction & fold
*View D - 0.75", 0.19" and 0.25"
*Section FF - typical butt seam

E5-19-5227
Floor, entrance
Section BB - 1.50"

E5-19-5248
Entrance, Shroud Assy
Check "D" ring alignment with airlock door (use note 13)

*Must be checked in process.

4.3.3.3 Inspection levels and AQLs. The inspection levels and AQLs expressed in terms of defects per hundred units, shall be as follows:

| Examination | Inspection level | AQL |
|---|---|---|
| For defects applicable table I | II | 40.0 Major<br>100.0 Major and minor defects combined |
| For defects applicable 4.3.3.2 | II | 6.5 |

4.3.3.4 End product testing. Each shelter/entrance assembly shall be tested in an inflated condition as specified in 4.4.5, 4.4.6 and 4.4.7. Testing as described in 4.4.5 shall precede all other testing. Any nonconformity shall be cause for rejection of the unit. Testing as specified in 4.4.8 shall be performed in accordance with inspection level S-1.



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4.3.4 Examination of packaging requirements. An examination shall be made to determine that preservation, unit packing, packing, and marking requirements are in compliance with section 5 of this specification. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units shall be 2.5. Examination for closure and strapping shall be in accordance with the applicable container specification.

TABLE III. Examination of preparation for delivery

| Item | Defect | Classification Major | Minor |
|---|---|---|---|
| Marking | (a) Omitted, illegible, or incorrect | X | |
| | (b) Improper size, location, sequence or method of application | | X |
| Materials | Any component missing or damaged | X | |
| Workmanship | Components lacking protection from abrasion or puncture | X | |
| Content | Not as specified | | |

4.4 Testing.

4.4.1 Breaking strength. The cut-strip breaking strength of cemented joints, seams, or splices shall be determined by method 5102 of Fed. Test Method Std. No. 191, except the rate of loading shall be at two inches per minute.

4.4.2 Dead load test of sewn web seam.

4.4.2.1 Specimen. The test specimen shall be two rectangles of fabric, one from the skin and one from the web. The skin fabric shall incorporate a properly sewn junction (see 3.3.1.1) in its approximate center, by which the web is attached (see figure 1). No thread breaks or back stitching permitted.

4.4.2.2 Method. The dead-weight loading apparatus to be employed shall conform in principle to the configuration illustrated in figure 2. Seams of test specimens shall successfully support a load of 50 lbs. per inch for a period of 24 hours at room temperature without any noticeable separation. The same type seam shall withstand a load of 25 lbs. per inch for a period of 24 hours at 180°F.



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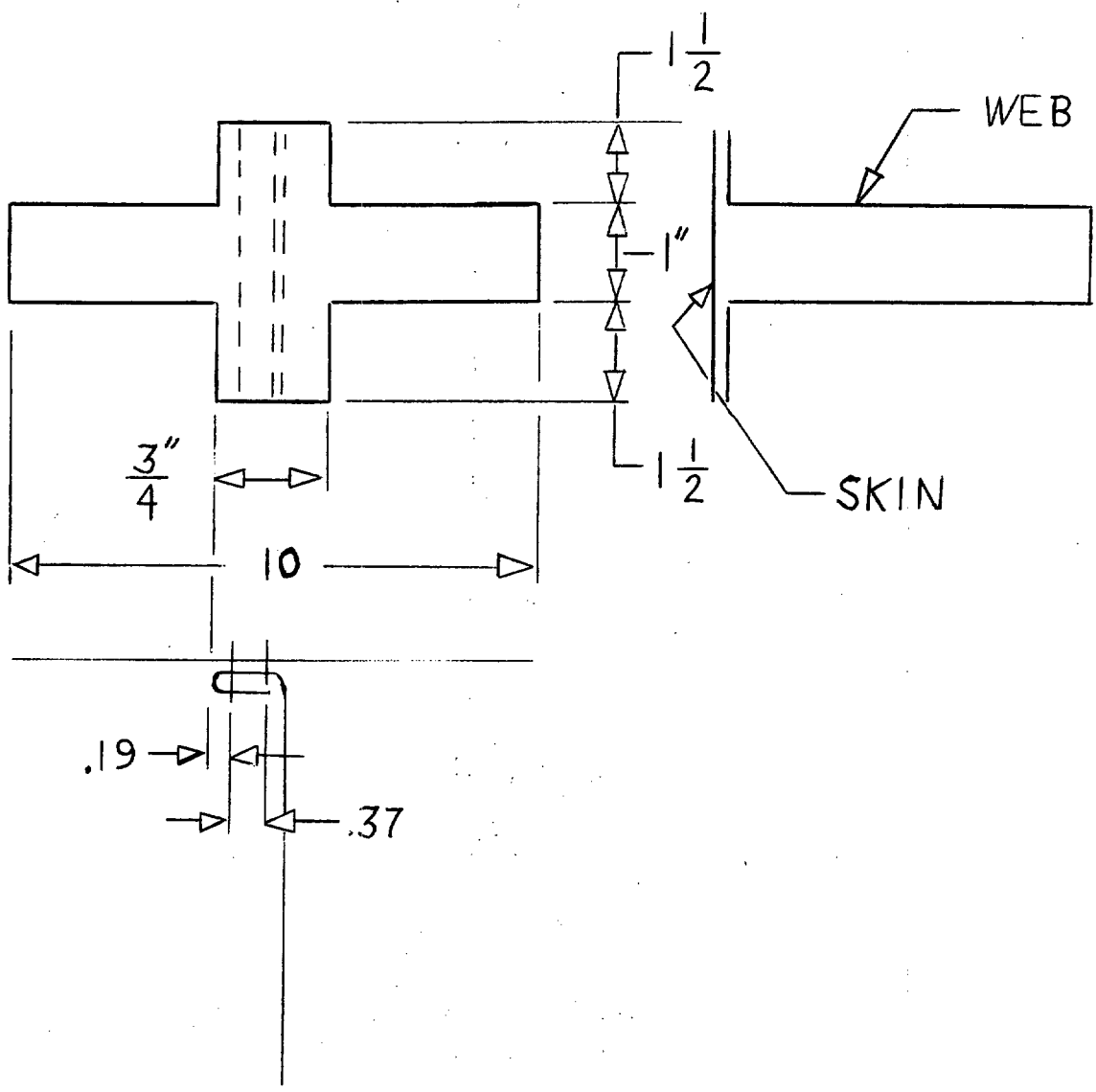



FIGURE 1. Sewn seam dead load sample.



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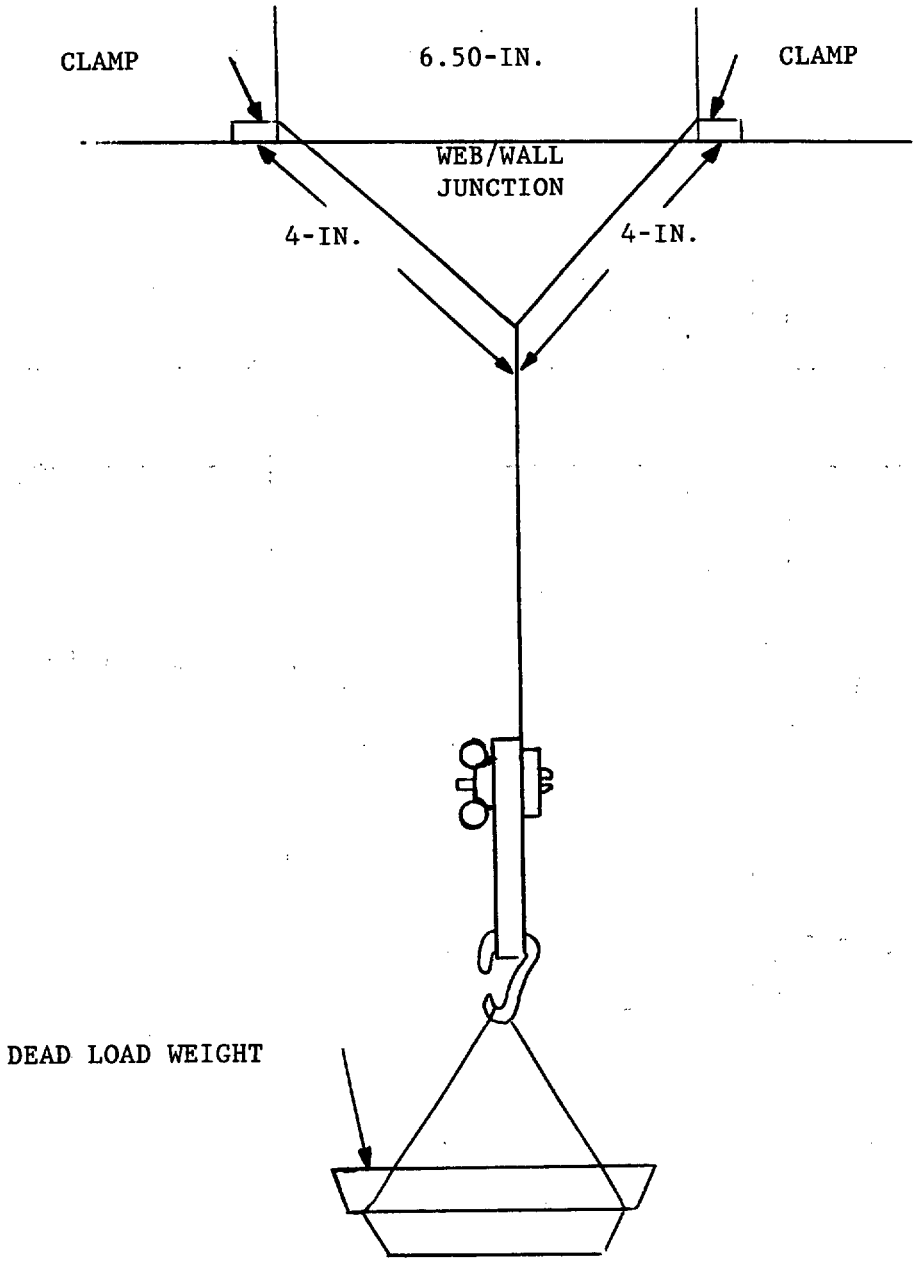



FIGURE 2. Dead-load apparatus



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4.4.3 Dead load tests of adhesives for butted seam.

4.4.3.1 Sample butt seam. The sample seam shall be one yard long with approximately 10 inches of wall fabric on both sides of the seam, and shall be prepared in the same manner as is to be used in production of shelter seams, so that the sample seam is representative of the shelter butted seam. Except that all three coated cloth parts of the sample seam for test shall have warp threads aligned at right angles to the run of the seam, the sample butt seam shall include the adhered sealing tape. The adhesives shall be those specified in 3.1.8. The sample seam shall be cut into specimen after seven days cure. The cure may be assisted for this test by subjecting the seam or specimens to heating for 1 hour at 170°F prior to application of the dead lock.

4.4.3.2 Test specimens. Test specimens for all tests shall be one inch wide and shall be cut across the long direction of the seam. The long sides of the specimen shall parallel the warp threads of the wall fabrics.

4.4.3.3 Dead load test conditions. The specimen prepared as in 4.4.3.2 shall be subjected to the dead load test described in 4.4.3.4 under each of the following conditions:

(a) 100 lbs: 4 hours, at room temperatures (70° to 90°F).

(b) 40 lbs: 4 hours, at 200°F ± 5°F.

(c) 100 lbs: 4 hours, at -25°F ± 2°F.

(d) 40 lbs: 30 days, at 86°F ± 2°F, 95 to 100% rh.

(e) 100 lbs: 4 hours, at room temperature (70° to 90°F) after 100 hours, in accelerated weathering machine without filters in accordance with method 5804 of Fed. Test Method Std. No. 191.

4.4.3.4 Dead load test equipment and procedure. For each test condition in 4.4.3.3, the specimen to be tested shall have one end anchored in a suitable grip and the specified force shall be applied continuously for the specified time at the other end of the specimen. The anchoring grip and the force-application grip shall be of a type which precludes failure of the specimen at or in the grips, and they shall be applied not closer than two inches from the butt strap. The grips shall hold the specimen uniformly across the one inch dimension and the force shall be applied uniformly without a twist. Suitable gage marks shall be made at the butt strap edges immediately after application of the test load to assist in determining the amount, if any, of relative slippage of the lap seam at the conclusion of the test. At the conclusion of testing under each specified condition, the specimen shall be examined for occurrence of slippage or separation of the structural butt seam and the occurrence of separation, edge peeling, and edge curling of the sealing tape.



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4.4.4 Dead load tests of cemented seams during production.

4.4.4.1 Sample butt seam and specimens. Sample butt seam and specimens shall be prepared as specified in 4.4.3.1 and 4.4.3.2 except that the seam shall be assembled concurrently with production of shelter seams using identical personnel, procedures, and materials so that the sample seam is representative of the shelter butted seam except that all three coated cloth parts of the sample seam for test shall have warp threads aligned at right angles to the run of the seam.

4.4.4.2 Test conditions. One specimen shall be subjected to the dead load test described in 4.4.3.4 under each of the following conditions:

(a) 100 lbs: 4 hours, at room temperatures (70° to 90°F).

(b) 40 lbs: 4 hours, at 200°F ± 5°F.

4.4.5 Overpressure seam strength. The shelter/entrance assembly ribs shall be inflated to a static pressure of 2.5 ± 0.1 psig pressure for a period of two hours. Upon expiration of that time there shall be no separation of the wall seams and no evidence of separation of the seams at the section junctures.

4.4.6 Air leakage.

4.4.6.1 Operational leak test. Use the centrifugal blower to initially inflate the shelter/entrance assembly ribs. Then transfer to a compressed air supply and maintain the shelter/assembly rib pressure at 42 inches water gage (iwg) using the apparatus as shown in figure 3. Make sure that all fittings are tight and free of leaks. At 42 iwg, the shelter/entrance assembly rib leakage shall not exceed 3.0 cfm.

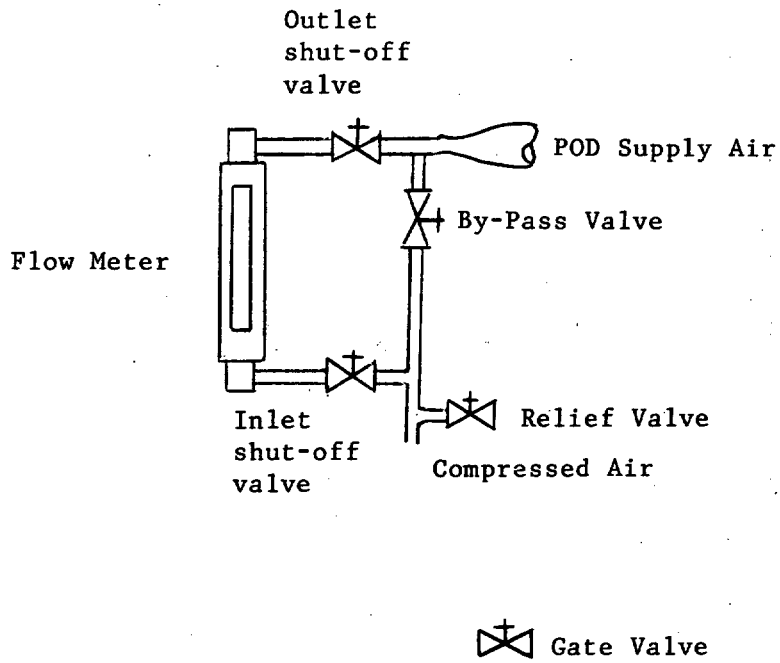
4.4.6.2 Stand-up leak test. A pressure tap shall be provided in a cover assembly for a dump valve. This tool shall be separate and capable of being fitted into any dump valve of the shelter or entrance where a pressure reading may be desired. The shelter/entrance ribs shall be inflated to a static pressure of 42 iwg. The inflation duct is to be securely clamped, sealing the shelter and airlock from the inflation fan. The shelter/entrance assembly shall remain erect for a period of 2 hours without additional inflation. At the end of that period the internal rib pressure shall be a minimum of 14 iwg in compliance with the requirements of 3.3.6.2.

4.4.7 Internal pressurization. Air flow at 150 ± 10 cfm at a maximum of 1.6 iwg shall be introduced through the recirculation duct and shall pressurize the interior of the shelter. The shelter interior pressure shall be not less than 0.8 iwg. The entrance interior pressure shall be not less than 0.3 iwg.

4.4.8 Interchangeability. Units shall have their entrances and shelters interchanged to demonstrate interchangeability and performance in accordance with 4.4.7.



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Flow Meter - Fischer & Porter or equivalent.

700 Series Flowrater, Tube Specification B4-21-10/70 SG

| Float Specification | SCFM Range |
|---|---|
| BS VT-45 | 2.48 |
| BS VT-44 | 3.4 |

FIGURE 3. Leak test.



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4.4.9 Seam and valve leakage. In the event that a unit does not conform to the tests of 4.4.5, 4.4.6, or 4.4.7, the unit shall be repaired as necessary and resubjected to the tests of 4.4.5, 4.4.6, and 4.4.7.

4.4.10 Agents penetration of window material and chloroprene-backed tape.

4.4.10.1 Initial agents penetration resistance. Resistance to permeation by Mustard (H) shall be determined as specified in method 204 and resistance to permeation by GB shall be determined as specified in method 206 of MIL-STD-282.

4.4.10.2 Low temperature test of tent window material. A 4 inches by 4 inches specimen shall be exposed for 60 ± 10 minutes at a temperature of -65°F ± 1.8°F and tested as specified in Method 5874 of Fed. Test Method Std. 191 except that the hydrostatic resistance shall not be determined. Samples shall be visually examined for cracking and delamination (if laminated) and subsequently tested for agents penetration as specified in 4.4.10.3.

4.4.10.3 Agents penetration resistance after low temperature for window material only. Specimens satisfactorily passing the visual examination after the low temperature test shall be subjected to agents penetration tests outlined in 4.4.10.1 above, with test specimens cut to include the crease line and with the agents contacting the crease line.

4.4.11 Haze. Haze shall be determined on three specimens according to ASTM D 1003 using procedure A. The average of three determinations shall be used to ascertain conformance to haze requirement of 3.1.3.

5. PACKAGING

5.1 Preservation.

5.1.1 Cleaning and drying. The shelter, entrance and accessory items (PL5-19-6822) shall be cleaned in accordance with cleaning method C-1 of MIL-P-116 and any applicable drying method listed therein.

5.1.2 Preservation application. The components of the shelter and entrance and accessory items susceptible to corrosion (nonpainted metal surfaces) shall be coated with P2 preservative of MIL-P-116, and damaged painted surfaces shall be treated and repainted in accordance with MIL-STD-171 except that the paint specified on the applicable drawing shall be used.

5.1.3 Cushioning and folding shelter and entrance assemblies.

5.1.3.1 Shelter assembly (E5-19-5178). Prior to folding, all sharp, protruding, or abrasive areas of the shelter assembly shall be cushioned with material conforming to type II, class B, minimum 3/16 inch thick of PPP-C-843. The cushioning material shall be secured with tape conforming to type IV of PPP-T-60. Special attention shall be given to (1) interconnecting fittings,



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(2) air-return outlet and air-supply inlet, (3) dump valves, and (4) tie-down rope assemblies. After the abrasive areas have been covered, the shelter assembly shall be folded, rolled, and secured as shown on Drawing E5-19-6823.

5.1.3.2 Entrance assembly (E5-19-5250). Prior to folding, cushion all sharp, protruding, or abrasive areas of the entrance assembly with material and secure with tape specified in 5.1.3.1 above. Special attention shall be given to (1) interconnecting fittings, (2) air-recirculating inlet, (3) air-recirculating outlet, (4) door assembly, (5) inflation valves, and (6) tie-down rope assemblies. After the abrasive areas have been covered, and with the entrance completely collapsed, fold as follows: (1) position entrance interior doors directly over the exterior doors, (2) lift one side of the entrance door frame and tuck (fold) protruding fabric within the door frame area, (3) position fabric on opposite side of entrance in the same manner, and (4) tuck side tie-down ropes in folds of fabric. Caution - Do not stand on folded entrance doors. Doors can be damaged.

5.1.4 Unit packing.

5.1.4.1 Accessory items. The (1) evacuation manifold, (2) repair kit inflated shelter, (3) zipper transition, and (4) pouch and cover assembly (see PL5-19-6822) shall be individually unit packed by method III of MIL-P-116. These four unit packs shall be consolidated in a weather-resistant fiberboard box conforming to type CF, variety SW, grade W5c, style FTC of PPP-B-636. The box shall have inside dimensions of 36 by 36 by 6 inches (length, width, and depth, respectively). Cushioning material conforming to PPP-C-843 shall be included as necessary to prevent movement of the packaged contents. The box shall be closed in accordance with the closure requirements for weather-resistant boxes in the appendix to PPP-B-636.

5.2 Commercial packaging (see 6.4). The shelter (E5-19-5178) and entrance (E5-19-5250) assemblies, cushioned and folded as specified in 5.1.3.1 and 5.1.3.2; and the accessory items (PL5-19-6822), unit packed as specified in 5.1.4.1, shall be prepared for shipment in accordance with the commercial packing requirements of ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs and the consolidated container (see 5.1.4.1) shall be marked in accordance with MIL-STD-129. A packing list shall be printed on, or pasted to the top surface of the consolidated container (see 5.1.4.1) and shall include the nomenclature and quantity of each part contained therein. The commercial shipping container (5.2) shall be marked in accordance with the marking requirements of ASTM D 3951.

5.3.1 Special marking. The shipping container, or containers (see 5.2) which contain the processed shelter assembly (see 5.1.3.1) and entrance assembly (see 5.1.3.2), shall be marked with the following:

FOR USE WITH M51 SHELTER SYSTEM (NSN 4240-00-854-4144)



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5.4 Repair parts. When the shelter (E5-19-5178) and entrance (E5-19-5250) assemblies; and applicable individual accessory items (PL 5-19-6822), are procured for storage and issue as a repair part, preservation, unit packing, packing, and marking shall be as specified on the applicable SPI, which is identified by the part number preceded by "P", or by the national stock number.

6. NOTES

6.1 Intended use. The shelter/entrance covered by this specification is a double walled, air inflated, chemical-biological (CB) shelter tent, capable of being erected by five men in 30 minutes. It is trailer-transportable, can be airdropped, and is designed to provide a CB hazard-free environment for field medical operations or in which personnel may eat, sleep, shave, perform other bodily functions and obtain relief from the physical and psychological stresses of wearing combat CB protective clothing. When erected, the shelter walls are inflated to 1.5 psi. Semicircular in cross section, the shelter has a maximum inside height of 7 feet 6 inches and a minimum floor area of 208 square feet (15 by 14 feet). The inside dimensions of the entrance are 11 feet long, 50 inches wide, and 80 inches high.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Requirement of a preproduction sample (see 3.4).
(c) Selection of the applicable levels of preservation (see 5.1 and applicable SPIs).
(d) When the shelter (PN 5-19-5178) or the entrance (PN 5-19-5250) are procured separately the following shall be Government furnished material (GFM).

Inflatable shelter - NSN 4240-00-152-1304.
Inflatable entrance - NSN 4240-00-152-1306.
Filter and utilities unit M68 - NSN 4240-00-152-1305.

6.3 Standard sample. For access to standard samples, address the acquisitioning office issuing the invitation for bids.

6.4 Commercial packing (see 5.2). Commercial packing may only be used when shipping the item from a vendor or supplier to a prime contractor or Government depot for further processing.

6.5 International interest. Certain provisions of this specification are the subject of International Stadnardization Agreement QSTAG-451. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.



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6.6 Changes from previous issue. Vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Air-supported shelter
Assembly, shelter-entrance
Chemical-biological protection
Chemical-biological shelter system
Inflatable entrance
Inflatable shelter
Protective shelter
Shelter system
Tent
Transportable protective shelter
Trailer-transportable shelter system

Preparing activity:

Army - EA

(Project No. 4240-A819)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-51422B(EA) | SHELTER AND ENTRANCE ASSEMBLY - CHEMICAL-BIOLOGICAL SHELTER SYSTEM: INFLATABLE |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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